

TELECOM ANALYTICS

Customer Churn Prediction

End-to-End Machine Learning & Explainable AI Project

IBM Telco Churn Dataset | Innovexis / Academic Project

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Dataset: IBM Telco Churn | 7,043 Records | 21 Variables

Python 3.12 | Scikit-learn | XGBoost | SHAP v0.48

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1. Project Overview & Business Context

Customer churn — the loss of a subscriber to a competitor or voluntary service cancellation — is one of the most costly challenges facing the telecommunications industry. According to the Harvard Business Review, **acquiring a new customer is anywhere from five to twenty-five times more expensive than retaining an existing one**. Every churned customer therefore represents lost recurring revenue plus a sunk acquisition cost that must be replaced at significant new expense.

This project develops an interpretable, end-to-end machine learning solution for a fictional telco company to proactively predict which customers are at risk of churning, enabling the business to intervene before revenue is lost.

Business Problem

- Companies invest substantial resources (marketing, discounts, onboarding costs) to acquire new customers.
- When a customer churns, both current and future revenue from that customer are lost permanently.
- Cost, time, and effort must be reinvested to replace each lost subscriber in the revenue base.
- Retaining existing customers is significantly more cost-effective than acquiring new ones.

Project Objectives

- Build a binary classification model to predict whether a customer will churn (Yes=1) or stay (No=0).
- Identify the most important features driving customer churn using feature importance and SHAP.
- Compare multiple machine learning algorithms on the same preprocessed dataset.
- Select and optimize the best-performing model using cross-validation and GridSearchCV tuning.
- Explain model predictions at both global and individual customer level using SHAP Explainability.
- Translate model findings into actionable, data-grounded business retention recommendations.

Evaluation Metric Strategy

The churn dataset is **imbalanced (~26.6% churners)**. ROC-AUC was chosen as the primary optimization metric for hyperparameter tuning since it measures discriminatory ability across all thresholds — more reliable than accuracy under class imbalance. Accuracy, Precision, Recall, and F1-Score provide the complete performance picture on the minority churn class.

2. Dataset Description

The **IBM Telco Customer Churn dataset** contains information about 7,043 customers of a fictional telecommunications company. Each record includes customer demographics, subscribed services, account information, and a binary churn label. Column names were converted from CamelCase to snake_case for Python convention compliance.

Variable (snake_case)	Type	Description
customer_id	Identifier	Unique customer ID — removed before modelling
churn	Target	Customer churned: Yes / No
gender	Categorical	Male or Female
senior_citizen	Binary	1=Senior citizen; 0=Not
partner	Categorical	Has a partner: Yes / No
dependents	Categorical	Has dependents: Yes / No
tenure	Numerical	Months the customer has used the service
phone_service	Categorical	Has phone service: Yes / No
multiple_lines	Categorical	Has multiple phone lines
internet_service	Categorical	Internet provider: DSL / Fiber optic / No
online_security	Categorical	Has online security add-on
online_backup	Categorical	Has online backup add-on
device_protection	Categorical	Has device protection add-on
tech_support	Categorical	Has tech support add-on
streaming_tv	Categorical	Streams TV via the service
streaming_movies	Categorical	Streams movies via the service
contract	Categorical	Contract term: Month-to-month / One year / Two year
paperless_billing	Categorical	Uses paperless billing: Yes / No
payment_method	Categorical	Payment method (4 types)
monthly_charges	Numerical	Monthly charge amount (\$)
total_charges	Numerical	Total charges over customer lifetime (\$)

Class	Count	Percentage
No Churn (0)	5,163	73.42%
Churn (1)	1,869	26.58%
Total	7,032	100% (post-cleaning)

The dataset is **moderately imbalanced** (73:27 ratio). ROC-AUC is prioritized over accuracy as the primary evaluation metric to properly assess model performance on the minority churn class.

3. Data Overview & Initial Quality Assessment

3.1 Dataset Dimensions & Structure

Metric	Value
Initial rows	7,043
Initial columns	21
Numerical variables	3 (senior_citizen, tenure, monthly_charges)
Categorical variables	17 (including customer_id, churn)
Duplicate records	0 — no duplicates found
Apparent missing values	0 — hidden as blank strings in total_charges

3.2 Hidden Data Quality Issue: total_charges

The **total_charges** column was stored as object (string) dtype instead of float64. Inspection revealed **11 rows containing blank strings (' ')** — customers with tenure=0 (brand-new accounts never billed). This prevented automatic numeric detection by pandas.

Issue	Column	Details	Action Taken
Incorrect dtype	total_charges	Stored as object; should be float64	pd.to_numeric(errors='coerce') — blanks become NaN
Hidden missing values	total_charges	11 blank strings converted to 11 NaN after coerce	Rows dropped (0.16% of data — minimal impact)
Naming convention	All columns	Original CamelCase format	Converted to snake_case for Python PEP8 compliance

3.3 Statistical Summary

Variable	Mean	Std Dev	Min	Median	Max
senior_citizen	0.162	0.369	0.00	0.00	1.00
tenure	32.42	24.55	1.00	29.00	72.00
monthly_charges	64.80	30.09	18.25	70.35	118.75
total_charges	2,283	2,267	18.80	1,397	8,684.80

4. Data Cleaning Pipeline

A systematic 5-step data cleaning pipeline was implemented to resolve all identified data quality issues before exploratory analysis and model development.

Step 1 — Convert total_charges to Numeric

The total_charges column contained 11 blank string entries that prevented numeric conversion. Using errors='coerce' replaced these blank strings with NaN and successfully converted the column from object to float64 dtype.

```
pd.to_numeric(df['total_charges'], errors='coerce')
```

Step 2 — Remove 11 Missing Rows

The 11 NaN rows (only 0.16% of data) were removed rather than imputed. These corresponded to customers with tenure=0 — brand-new accounts with no billing history. Dropping was preferred over imputation (e.g., \$0 or median) to avoid introducing artificial values for customers who had never been charged.

```
df = df.dropna(subset=['total_charges']) → Final shape: (7,032 x 21)
```

Step 3 — Verify Duplicates

No duplicate records were found in the dataset. This step confirmed data integrity and ensured no observations would bias the training process through repetition.

```
df.duplicated().sum() → 0 duplicate records
```

Step 4 — Validate Categorical Values

All categorical variables were inspected for unexpected values, typos, encoding errors, or inconsistencies. No invalid entries were found. Categories 'No internet service' and 'No phone service' were identified for simplification in the feature engineering stage.

```
df[column].value_counts() for all 17 categorical columns
```

Step 5 — Save Cleaned Dataset

The fully cleaned dataset was saved to disk for reproducibility and to enable reuse across analysis stages without repeating the cleaning pipeline.

```
df.to_csv('telco_churn_cleaned.csv', index=False)
```

5. Exploratory Data Analysis (EDA)

EDA was conducted through 23+ structured visualizations using a combination of univariate, bivariate, and correlation analyses to understand feature distributions and their relationship with customer churn.

5.1 Univariate Analysis

Chart 1 — Target Variable Distribution (Count Plot + Percentage)

Insight: 73.42% of customers did not churn; 26.58% churned. The moderate imbalance confirms that ROC-AUC and F1-Score are more appropriate metrics than raw accuracy.

Charts 2-4 — Numerical Variable Distributions (Histogram + KDE: Tenure, Monthly Charges, Total Charges)

Insight: Tenure shows a near-bimodal distribution with peaks at very short tenures (<6 months) and long tenures (>60 months), revealing two distinct segments: new at-risk customers and long-term loyal customers. Monthly charges are right-skewed, concentrated in the \$60-\$100 range. Total charges are strongly right-skewed, reflecting tenure accumulation effects.

Charts 5-19 — Categorical Variable Count Plots (15 variables)

Insight: Majority of customers have phone service. Fiber optic is the most common internet type. Month-to-month contracts dominate the customer base. A significant share of customers do not subscribe to value-added services (online security, tech support, device protection).

5.2 Bivariate Analysis — Churn by Feature

Chart 20 — Gender vs Churn

Insight: Churn rates are nearly equal between male and female customers. Gender is not a meaningful predictor of churn — service dissatisfaction drivers are independent of gender demographics.

Chart 21 — Senior Citizen vs Churn

Insight: Senior citizens exhibit a disproportionately high churn rate relative to their population share (~16% of customers). They represent a high-priority retention segment requiring dedicated attention.

Chart 22 — Contract Type vs Churn

Insight: Month-to-month contract customers show dramatically higher churn numbers. Two-year contract customers have the lowest churn rate — the contractual commitment creates a powerful retention mechanism.

Chart 23 — Tenure vs Churn (Box Plot)

Insight: Churned customers have significantly lower median tenure (~10 months) vs. retained customers (~38 months). The first 12 months represent the highest-risk churn period for any new customer.

Chart 24 — Monthly Charges vs Churn (Box Plot)

Insight: Customers who churned have higher median monthly charges (~\$80) vs. non-churners (~\$61). High monthly costs are a significant churn driver — customers who perceive poor value relative to cost are more likely to leave.

Chart 25 — Total Charges vs Churn (Box Plot)

Insight: Retained customers have substantially higher total charges, primarily reflecting their longer tenure. Total charges serve mainly as a tenure proxy in this dataset.

5.3 Correlation Analysis (Heatmap)

- **tenure and total_charges** are strongly positively correlated — longer-tenured customers have naturally accumulated higher total bills.
- **monthly_charges and total_charges** show moderate positive correlation, expected given the billing relationship.
- **senior_citizen** shows very weak correlations with financial variables, confirming its independent demographic predictive value.
- No severe multicollinearity was detected that would destabilize model training.

5.4 Key Business Insights Summary

EDA Finding	Business Implication
27% churn rate	Significant revenue at risk — proactive retention is clearly cost-justified
Month-to-month contract dominance	Contract migration to annual plans is the highest-leverage retention lever
New customers (<6 months) at risk	Onboarding and early engagement programs are operationally critical
High monthly charges correlate with churn	Pricing review and value-perception improvements are needed
Senior citizens churn disproportionately	Dedicated senior support programs and pricing tiers are warranted
No online security/tech support -> higher churn	Value-added service bundling improves customer stickiness
Two-year contracts -> near-zero churn	Long-term contract incentives have a strong measurable retention effect

6. Feature Engineering & Data Preparation

The cleaned dataset was transformed into a machine-learning-ready format through a structured 6-step feature engineering pipeline before model training.

Step 1 — Encode Target Variable

The target variable churn was mapped from string to binary integers: No=0 (Retained) and Yes=1 (Churned). This is required for all scikit-learn classifiers and enables probability calibration.

```
df_ml['churn'].map({'No': 0, 'Yes': 1})
```

Step 2 — Simplify Categorical Values

Six internet add-on columns (online_security, online_backup, device_protection, tech_support, streaming_tv, streaming_movies) contained 'No internet service' — logically equivalent to 'No' for customers without internet. Similarly, multiple_lines contained 'No phone service'. Replacing these reduces unnecessary categories and prevents spurious dummy variable creation during encoding.

```
Replace 'No internet service' with 'No' (6 columns) | Replace 'No phone service' with 'No' (multiple_lines)
```

Step 3 — Drop customer_id

The customer ID is a unique alphanumeric identifier with zero predictive value. Removing it prevents the model from memorizing individual IDs and ensures all model learning comes from business-relevant features only.

```
df_ml.drop(columns='customer_id', inplace=True)
```

Step 4 — One-Hot Encoding

All remaining categorical variables were one-hot encoded into binary dummy variables. drop_first=True avoids the Dummy Variable Trap (perfect multicollinearity). The final encoded dataset has 23 feature columns and 1 target column.

```
pd.get_dummies(df_ml, drop_first=True, dtype=int) -> Shape: (7,032 x 24)
```

Step 5 — Train-Test Split

Stratified sampling (stratify=y) ensures the churn proportion (~26.6%) is preserved identically in both training and testing sets. Without stratification, the test set could be unrepresentative of the actual churn rate.

```
train_test_split(X, y, test_size=0.20, random_state=42, stratify=y) | Train: 5,625 | Test: 1,407
```

Step 6 — Selective Feature Scaling

Only the three continuous numerical variables were standardized (Mean=0, Std=1). Binary one-hot encoded dummy variables do not benefit from scaling. The scaler was fitted exclusively on training data and applied transform-only to the test set, preventing data leakage. Tree-based models received unscaled data; Logistic Regression received scaled data.

```
StandardScaler applied to: tenure, monthly_charges, total_charges only
```

7. Baseline Machine Learning Models

Three supervised classification algorithms were trained and evaluated on the preprocessed Telco churn dataset. Each model was assessed using five metrics, a Confusion Matrix, ROC Curve, and Feature Importance visualization.

7.1 Logistic Regression

Configuration: random_state=42 | Trained on StandardScaler-normalized data. Logistic Regression is the standard interpretable baseline for binary classification — coefficients can be directly interpreted as log-odds of churn for each feature.

Accuracy	Precision	Recall	F1-Score	ROC-AUC
80.31%	64.83%	56.68%	60.49%	83.62%

Class	Precision	Recall	F1-Score	Support
Non-Churn (0)	0.85	0.89	0.87	1,033
Churn (1)	0.65	0.57	0.60	374

Analysis: Logistic Regression achieves the highest ROC-AUC at baseline (83.62%) and leads on accuracy (80.31%). Recall of 56.68% means it correctly flags slightly more than half of all churners. As a linear model, it may not capture complex non-linear interactions between contract type, tenure, and pricing variables.

7.2 Random Forest Classifier

Configuration: n_estimators=100 (default), random_state=42 | No feature scaling required. Combines 100 decision trees using bagging to reduce variance and capture non-linear patterns.

Accuracy	Precision	Recall	F1-Score	ROC-AUC
78.96%	63.00%	50.53%	56.08%	81.84%

Class	Precision	Recall	F1-Score	Support
Non-Churn (0)	0.83	0.89	0.86	1,033
Churn (1)	0.63	0.51	0.56	374

Analysis: Random Forest achieves 78.96% accuracy with the lowest Recall at baseline (50.53%) — correctly identifying only half of all churners. The feature importance plot confirms tenure, monthly_charges, and total_charges as the top predictors, consistent with EDA findings.

7.3 XGBoost Classifier

Configuration: random_state=42, eval_metric='logloss' | No scaling required. Sequential gradient boosting with built-in regularization.

Accuracy	Precision	Recall	F1-Score	ROC-AUC
77.83%	58.91%	54.81%	56.79%	81.97%

Class	Precision	Recall	F1-Score	Support
Non-Churn (0)	0.84	0.86	0.85	1,033
Churn (1)	0.59	0.55	0.57	374

Analysis: XGBoost achieves the highest Recall among baseline models (54.81%), correctly identifying more churners than Random Forest despite slightly lower accuracy. Its gradient boosting architecture naturally handles the minority

churn class better at baseline.

7.4 Baseline Model Comparison

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	80.31%	64.83%	56.68%	60.49%	83.62%
Random Forest	78.96%	63.00%	50.53%	56.08%	81.84%
XGBoost	77.83%	58.91%	54.81%	56.79%	81.97%

Baseline Ranking by ROC-AUC: Logistic Regression (83.62%) > XGBoost (81.97%) > Random Forest (81.84%).
Logistic Regression leads at baseline, with both ensemble methods delivering similar overall performance.
Hyperparameter tuning is expected to improve all three models.

8. Hyperparameter Tuning

All three models were systematically tuned using **GridSearchCV with 5-fold cross-validation**. **ROC-AUC** was used as the primary scoring metric throughout — it measures the model's overall ability to separate churners from non-churners across all classification thresholds, making it more informative than accuracy under class imbalance.

8.1 Logistic Regression — GridSearchCV

Hyperparameter	Grid	Best Value
C (Regularization)	[0.01, 0.1, 1, 10, 100]	C=10
penalty	['l2']	l2
solver	['lbfgs']	lbfgs

Best CV ROC-AUC: 0.8461 | Finding: C=10 (lower regularization) allowed the model to fit more complex patterns in the high-dimensional one-hot encoded feature space.

8.2 Random Forest — GridSearchCV

Hyperparameter	Grid	Best Value
n_estimators	[100, 200]	200
max_depth	[5, 10, 20]	5
min_samples_split	[2, 5]	2
min_samples_leaf	[1, 2]	1

Best CV ROC-AUC: 0.8451 | Finding: Shallow trees (max_depth=5) with more estimators (200) reduced overfitting and improved cross-validation performance, though test Recall decreased significantly.

8.3 XGBoost — GridSearchCV

Hyperparameter	Grid	Best Value
n_estimators	[100, 200]	100
learning_rate	[0.01, 0.1]	0.1
max_depth	[3, 5]	3
subsample	[0.8, 1.0]	0.8
colsample_bytree	[0.8, 1.0]	0.8

Best CV ROC-AUC: 0.8487 — HIGHEST among all tuned models | Shallow trees (depth=3) with subsampling (0.8) prevent overfitting. Learning rate of 0.1 provides a good speed-performance trade-off.

8.4 Cross-Validation ROC-AUC Summary

Model	Best CV ROC-AUC	Rank
XGBoost	0.8487	#1 — Selected as Final Model
Logistic Regression	0.8461	#2
Random Forest	0.8451	#3

XGBoost achieved the highest cross-validation ROC-AUC (0.8487), confirming its selection as the final production model.

9. Final Model Selection & Comparison

9.1 Tuned Model Test Set Performance

Model	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Tuned Random Forest	78.82%	66.38%	41.18%	50.83%	83.94%
Tuned XGBoost	79.60%	63.72%	54.01%	58.47%	83.91%
Tuned Logistic Regression	80.10%	64.07%	57.22%	60.45%	83.53%

9.2 Final Model: Tuned XGBoost

FINAL MODEL: Tuned XGBoost Accuracy: 79.60% | Precision: 63.72% | Recall: 54.01% | F1-Score: 58.47% | CV ROC-AUC: 84.87% | Test ROC-AUC: 83.91%

Class	Precision	Recall	F1-Score	Support
Non-Churn (0)	0.84	0.89	0.86	1,033
Churn (1)	0.64	0.54	0.58	374

9.3 Model Selection Rationale

- Highest CV ROC-AUC (0.8487):** Strongest discriminatory ability during cross-validation — the most reliable generalization performance indicator.
- Best Recall among tree models (54.01%):** XGBoost correctly identifies more churners than Random Forest. Missing a churner (False Negative) costs the company a full customer relationship; a false alarm (False Positive) only costs a retention offer.
- Best F1-Score among ensemble models (58.47%):** Strongest balance between precision and recall on the imbalanced churn class.
- Tuned Random Forest Disqualification:** Despite competitive ROC-AUC (83.94%), tuned Random Forest's Recall collapsed to 41.18% — it misses nearly 60% of all churners. This makes it unsuitable despite its high precision.
- Logistic Regression Alternative:** Tuned Logistic Regression achieves the highest Recall (57.22%) but lower CV ROC-AUC (0.8461) and limited non-linear pattern capture compared to XGBoost.

9.4 Baseline vs Tuned Comparison

Model	Stage	Recall	F1-Score	ROC-AUC
Logistic Regression	Baseline	56.68%	60.49%	83.62%
Logistic Regression	Tuned	57.22%	60.45%	83.53%
Random Forest	Baseline	50.53%	56.08%	81.84%
Random Forest	Tuned	41.18%	50.83%	83.94%
XGBoost	Baseline	54.81%	56.79%	81.97%
XGBoost	Tuned	54.01%	58.47%	83.91%

Key Insight: Tuning for ROC-AUC improved overall discrimination for all models. However, Random Forest's Recall dropped significantly post-tuning — a known trade-off when optimizing for threshold-independent metrics. XGBoost maintained high Recall while improving ROC-AUC, confirming its suitability as the final model.

10. Explainable AI using SHAP

The tuned XGBoost model, while delivering strong predictive performance, is a 'black-box' whose internal decision logic is not directly interpretable by business stakeholders. **SHAP (SHapley Additive exPlanations) v0.48** was applied to explain model predictions at both global and individual customer levels using a shap.Explainer (TreeExplainer internally for XGBoost).

10.1 Why SHAP?

- SHAP is grounded in cooperative game theory (Shapley values) — each feature receives a fair attribution for its marginal contribution to the prediction.
- Unlike Gini feature importance, SHAP shows both the **magnitude** (how much impact) and **direction** (toward churn or away from churn) of each feature.
- SHAP enables regulatory and ethical compliance — companies offering targeted retention discounts must be able to justify why specific customers were flagged as high-risk.
- SHAP local explanations empower customer success teams to personalize retention conversations based on each individual customer's specific risk profile.

10.2 Global Feature Importance (SHAP Ranking)

Rank	Feature	Mean SHAP	Churn Risk Direction
1	tenure	0.5461	Low tenure (new customers) -> High churn risk
2	contract_Two year	0.4945	Two-year contract -> Strongly reduces churn
3	internet_service_Fiber optic	0.3529	Fiber optic -> Associated with higher churn
4	contract_One year	0.2316	One-year contract -> Moderate churn reduction
5	payment_method_Electronic check	0.1955	Electronic check -> Higher churn risk
6	monthly_charges	0.1741	Higher charges -> Higher churn risk
7	internet_service_No	0.1535	No internet -> Lower churn (fewer at-risk services)
8	total_charges	0.1511	Higher total -> Lower churn (tenure proxy)
9	paperless_billing_Yes	0.1145	Paperless billing -> Slight churn increase
10	online_security_Yes	0.1096	Online security subscription -> Reduces churn
11	tech_support_Yes	0.1003	Tech support subscription -> Reduces churn
12	multiple_lines_Yes	0.1143	Multiple lines -> Slight churn association

10.3 SHAP Visualization Outputs

Beeswarm Plot (SHAP Summary)

Displays each test customer as a dot for each top feature, colored by feature value (red=high, blue=low). Tenure appears at the top — low-tenure customers (blue dots on the right) have strongly positive SHAP values, pushing predictions toward churn. Two-year contract (red=two-year) has strongly negative SHAP values.

SHAP Bar Plot (Global Importance)

Ranks all 23 features by mean absolute SHAP value. Confirms tenure and contract_Two year as the dominant churn predictors, with internet_service_Fiber optic in third position.

Waterfall Plot — Customer #0 (Local Explanation)

Shows the feature-by-feature breakdown of why the model predicted this specific customer's churn probability. Starting from the baseline expected value, each feature contributes a positive (red, toward churn) or negative (blue, away from

churn) SHAP value.

Force Plot — Customer #0

Visualizes the same individual prediction as a horizontal force diagram — red arrows push toward churn, blue arrows push away. Enables rapid visual communication of risk factors to non-technical business stakeholders.

SHAP Dependence Plot — Tenure

Scatter plot showing how the SHAP value for tenure changes across its full range. Tenure below 12 months generates strongly positive SHAP values (high churn risk); tenure above 36 months generates consistently negative SHAP values (strong retention signal). The non-linearity confirms the critical early engagement window.

Waterfall Plot — Customer #25

Demonstrates that the same feature can have opposite SHAP contributions for different customers, confirming the model's individualized (not rule-based) risk assessment capability.

10.4 SHAP Business Takeaways

- A customer with tenure <6 months on a month-to-month contract with fiber optic internet paying by electronic check represents the **highest-risk churn profile** in the dataset.
- Fiber optic customers churn more — possibly because fiber plans carry higher monthly charges, worsening perceived cost-value ratio, especially for new customers.
- Electronic check payers show higher churn — potentially reflecting lower engagement and loyalty compared to automatic payment customers who have more inertia.
- Online security and tech support subscriptions create meaningful protective effects against churn — these services increase switching friction and improve perceived value.
- Two-year contract commitment provides the second-strongest churn protection after tenure — nearly equal in SHAP importance to the customer's tenure duration.

11. Business Recommendations

Based on the combined findings from EDA, model results, and SHAP explainability, the following 8 data-driven retention strategies are recommended for immediate implementation.

- **1. Contract Upgrade Campaign (Highest Priority)**

The single most impactful retention lever is migrating month-to-month customers to longer-term contracts. Offer meaningful incentives — 15-25% first-year discounts, free service upgrades, or loyalty credits — specifically targeted at customers predicted at high churn risk. SHAP confirms that two-year contracts are the second-strongest churn reducer after tenure.

- **2. New Customer Onboarding Program (Months 1-12)**

Tenure is the #1 churn predictor (SHAP: 0.546). Customers in their first 12 months carry the highest churn risk. Implement structured onboarding: proactive check-ins at months 1, 3, and 6; a dedicated new-customer success line; and early satisfaction surveys to catch issues before they trigger a cancellation decision.

- **3. Fiber Optic Value Perception Strategy**

Fiber optic internet is the 3rd-most important churn driver. Conduct competitive pricing analysis for fiber plans and introduce value bundling (streaming + security + support) that justifies the higher monthly cost. Tiered fiber pricing can address price-sensitive customers without losing premium-segment revenue.

- **4. Electronic Check to Auto-Payment Migration**

Electronic check payment correlates with higher churn (SHAP: 0.196). Offer a small monthly discount (\$2-\$5) for customers who switch to automatic credit card or bank transfer payments. This improves both retention and collections reliability simultaneously.

- **5. Value-Added Service Bundling**

Online security and tech support subscriptions significantly reduce churn. Create introductory bundle offers at 50% reduced cost for the first 3 months, giving customers the opportunity to experience their value before committing to full pricing. Once adopted, these services create switching friction.

- **6. Senior Citizen Dedicated Support Tier**

Senior citizens exhibit disproportionately high churn. Create a senior-specific service tier with simplified billing, dedicated phone-based support, in-home technical assistance options, and a senior loyalty discount to improve retention within this underserved segment.

- **7. High Monthly Charge Proactive Review**

Customers with monthly charges above \$80 are at elevated churn risk. Implement a monthly billing review program where account managers proactively reach out to high-charge customers with personalized value reviews, plan optimization recommendations, and loyalty retention offers before dissatisfaction escalates.

- **8. Real-Time CRM-Integrated Churn Scoring**

Deploy the tuned XGBoost model as a REST API integrated with the CRM system, generating a monthly churn probability score for every active customer. Customers with predicted probability >0.45 are automatically flagged for targeted retention outreach, with SHAP local explanations surfaced in the CRM dashboard to personalize each retention conversation.

12. Conclusion & Future Scope

This project delivered a complete, production-quality end-to-end machine learning pipeline for telecom customer churn prediction. Starting from raw customer data, every stage of the data science lifecycle was addressed systematically:

- Data quality assessment including hidden blank-string detection in total_charges column.
- 23+ chart EDA framework revealing behavioral, contractual, and demographic churn drivers.
- 6-step feature engineering pipeline: category simplification, one-hot encoding, stratified train-test split, and selective StandardScaler application.
- Development and evaluation of three baseline classifiers with Confusion Matrices, ROC Curves, and Feature Importance visualizations.
- GridSearchCV hyperparameter tuning for all three models with 5-fold cross-validation and ROC-AUC as the primary optimization metric.
- Final model selection: Tuned XGBoost | CV ROC-AUC=84.87% | Test ROC-AUC=83.91% | Recall=54.01% | F1=58.47%.
- SHAP Explainability: beeswarm plots, bar plots, waterfall plots, force plots, and scatter dependence plots.
- 8 actionable, data-grounded business recommendations for the customer retention team.

Final Model: Tuned XGBoost | CV ROC-AUC: 84.87% | Test ROC-AUC: 83.91% Correctly identifies 54% of at-risk customers before they churn, enabling proactive retention interventions that significantly reduce the 5-25x higher cost of new customer acquisition.

Future Scope

- **SMOTE Oversampling:** Apply Synthetic Minority Oversampling Technique to boost Recall for the minority churn class without sacrificing precision on the majority non-churn class.
- **Decision Threshold Optimization:** Lower the classification threshold from 0.50 to 0.35-0.40 to increase Recall based on a formal cost-benefit analysis of False Negative vs False Positive costs.
- **Customer Lifetime Value Integration:** Combine churn probability with estimated CLV to prioritize retention spend on high-value at-risk customers rather than treating all churners equally.
- **Survival Analysis:** Model time-to-churn using Kaplan-Meier or Cox PH models to predict not just whether a customer will churn, but when — enabling time-sensitive retention interventions.
- **A/B Testing Framework:** Use the model to assign customers to control and treatment groups, measuring the incremental retention impact of different intervention strategies empirically.
- **Deep Learning Models:** Explore TabNet or feedforward neural networks for potentially higher accuracy and better automatic feature interaction learning on the structured Telco feature set.
- **Real-Time API Deployment:** Deploy as a FastAPI REST endpoint returning churn probability + SHAP local explanations per customer, fully integrated into the CRM retention workflow.